

SANDHYA BARAL

077bct076.sandhya@pcampus.edu.np | [LinkedIn](#) | [GitHub](#) | [Personal Website](#)

PERSONAL STATEMENT

AI researcher driving software engineering and database expertise with advanced work in computer vision, NLP, and privacy-preserving machine learning. An award-winning academic record and international publications (IAPR, ISBM), demonstrating a proven ability to translate complex research into scalable systems in the domain of finance, documents and languages.

EDUCATION

Bachelor of Engineering in Computer Engineering

Pulchowk Campus, Institute of Engineering, Tribhuvan University, Nepal

Undergraduate

October 2021 – June 2025

- **Academic Standing:** 79.22%
- **Relevant Coursework:** Applied Mathematics (Linear Algebra, Calculus), Probability & Statistics, Data Structures & Algorithms, Database Management Systems, Operating Systems, Artificial Intelligence, Computer Networks, Object-Oriented Programming, Embedded Systems, Digital Signal Analysis and Processing.
- **Capstone Topic:** Handwritten Devanagari Character Recognition: Dataset Creation and Benchmarking with Transfer Learning
Supervisor: [Prof. Sanjeeb Prasad Pandey](#)

PUBLICATIONS

- Tiwari, S.*, Khadka, S.*, **Baral, S.***, & Panday, S. P. (2026, September). Handwritten Devanagari Character Recognition: Conjunct Consonants Dataset Creation and Benchmarking with Transfer Learning. In **17th IAPR International Workshop on Document Analysis Systems (DAS 2026)**. [[Paper](#), [Code](#)]
- Khadka, S., **Baral, S.**, Tiwari, S., & Ghimire, S. (2025, November). Character Recognition of Nepali Number Plate. In **Information Systems for Intelligent Systems**, Lecture Notes in Networks and Systems, vol. 1744. Springer. [[Paper](#), [Code](#)]

RESEARCH EXPERIENCE

Handwritten Devanagari Character Recognition

Capstone Research Project | Pulchowk Campus, IOE, Tribhuvan University | [[Code](#)]

May 2024 – April 2025

Abstract: This capstone research pioneered a novel, publicly releasable **annotated benchmark dataset for handwritten Devanagari conjunct consonants**, significantly extending existing character taxonomies and enabling richer linguistic modeling for a historically low-resource script. By pairing advanced preprocessing and systematic data-augmentation strategies with transfer learning across deep Convolutional Neural Network (CNN) architectures, including ResNet, VGG, and EfficientNet, the study achieved state-of-the-art (SOTA) recognition accuracy. This work has been accepted for presentation at the **17th IAPR Workshop on Document Analysis Systems (DAS)**, Vienna, Austria, 2026.

Supervisor: [Prof. Sanjeeb Prasad Pandey](#)

Abstract: This applied research introduced a robust **two-stage recognition pipeline for Nepali vehicle license plates**, integrating YOLOv8 for real-time, high-precision plate localization with fine-tuned deep CNNs for Devanagari script character classification. Trained on a custom-curated and annotated real-world dataset, the system leveraged transfer learning to adapt pre-trained networks, demonstrating consistent, high-accuracy performance across challenging environmental conditions such as severe illumination shifts, acute angles, and partial occlusions. The findings were presented in the **Information Systems for Intelligent Systems** and published in a **Springer volume**, and the project was awarded **Top Student** recognition by the Samsung Innovation Campus (2025).

Supervisor: [Prof. Sharad Kumar Ghimire](#)

PROJECTS

ArtiFuse: Intelligent Image Inpainting via Generative Deep Learning March 2024
Generative AI | [\[Code\]](#)

An end-to-end image inpainting system using **Pix2Pix-based conditional GANs**, integrating real-time web-based inference and human-in-the-loop interaction to achieve high-fidelity reconstruction of corrupted image regions.

Left-knee Segmentation January 2024
Medical Imaging | [\[Code\]](#)

Bone segmentation for left-knee CT scan images using classical image processing techniques, including thresholding, connected-component analysis, artifact removal through shape-based filtering, and automatic femur–tibia separation using anatomical cross-sectional profile analysis.

Support Vector Machine (SVM) from Scratch December 2023
Artificial Intelligence | [\[Code\]](#)

A beginner-friendly, from-scratch implementation of Support Vector Machines using NumPy, focused on the Iris dataset, serving as an educational guide for understanding SVM mathematics and optimization.

PROFESSIONAL EXPERIENCE

Machine Learning Researcher June 2025 – Present
Digital Age Nepal, Kathmandu, Nepal

- Research into privacy-preserving machine learning and model interpretability, maintaining data security and regulatory compliance while handling sensitive financial information.
- Optimized state-of-the-art (SOTA) graph networks and unsupervised learning algorithms to maximize predictive performance within strict computational and resource constraints.
- Engineered predictive time-series models for temporal anomaly detection, extracting behavioral insights from large-scale, high-cardinality relational financial databases.
- Designed automated information retrieval pipelines for complex legal literature, utilizing Optical Character Recognition (OCR) and Convolutional Neural Networks (CNNs) for accurate document data extraction and processing.

AWARDS & SCHOLARSHIPS

- 2025
- Top Student Award — Applied Data Science Course, Awarded by Samsung Innovation Campus, for the project “Character Recognition of Nepali Vehicle Number Plate” at Pulchowk Campus, Tribhuvan University.
- 2025
- Best Capstone Project Award — “Handwritten Devanagari Character Recognition: Conjunct Consonants Dataset Creation and Benchmarking with Transfer Learning,” Pulchowk Campus, Tribhuvan University.
- 2024
- Scholarship Recipient — 10-day (lectures + lab sessions + hackathon project) Fifth Annual Nepal AI School (ANAIS 2024), Nepal Applied Mathematics and Informatics Institute for Research (NAAMII).

TEACHING EXPERIENCE

- 2025
- Internet Literacy & Awareness (CIT) , Instructor | LOCUS Nationwide Campaign
- 2025
- Data Science Fundamentals , Instructor | LOCUS, Pulchowk Campus
- 2024
- Advanced Programming Concepts in C , Instructor | IT Club, Pulchowk Campus
- 2024
- React JS Intensive , Mentor | Girls To Code Bootcamp

TECHNICAL SKILLS

Programming Languages	Python, C/C++, HTML/CSS, SQL, MATLAB
ML / DL Frameworks	PyTorch, Keras
Libraries & Tools	OpenCV, Scikit-learn, NumPy, Pandas, Matplotlib, SciPy
Tools & Infrastructure	FastAPI, Blender, Linux, Git, Jupyter Notebook, L ^A T _E X
Data & Evaluation	Data Visualization, Data Mining, Statistical Analysis, Benchmarking, Dataset Curation & Annotation
Deep Learning	Computer Vision (2-D), Deep Learning (CNN, RNNs, Generative AI, YOLO), Pattern Recognition, Representation Learning, Prompt Learning, Image Segmentation Transformers, Transfer Learning, Anomaly Detection